

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 17 — Carbonyl Compounds (Aldehydes & Ketones)

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	17.1 Aldehydes & Ketones · Nucleophilic Addition (HCN) · Reduction (NaBH ₄) · Diagnostic Tests (2,4-DNPH, Brady's reagent)
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	267 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

17.1 Structure & Nucleophilic Addition

— Polarity of carbonyl C=O, nucleophilic addition of HCN via planar trigonal carbonyl carbon; stereochemistry

17.1 Reduction & Interconversions

— Hydride transfer mechanism using NaBH₄ (in aqueous ethanol) reducing aldehydes to 1° alcohols

17.1 Diagnostic Chemical Tests & Iodoform Reaction

— Condensation with 2,4-DNPH (Brady's reagent) and derivative melting points; distinction between aldehydes and ketones

SUBTOPIC 17.1 — CARBONYL COMPOUNDS (ALDEHYDES & KETONES)

1 Nucleophilic Addition Mechanism of HCN — 9701/22/M/J/23/Q5(a)–(d)

HARD · Syllabus 17.1

Aldehydes and ketones react with hydrogen cyanide in the presence of a trace of base or sodium cyanide via a nucleophilic addition mechanism. Fig. 1.1 illustrates the two-step pathway and the stereochemical consequences of attacking a planar carbonyl centre.

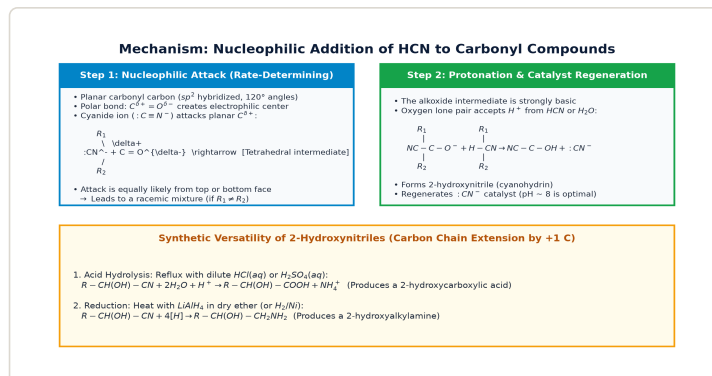


Fig. 1.1: Nucleophilic addition mechanism of HCN to propanal and subsequent synthetic transformations.

- ((a)) Draw the complete two-step mechanism for the reaction between propanal and cyanide ions ($:CN^-$). Include all relevant dipoles, lone pairs, curly arrows, and the structure of the intermediate. [4]

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- ((b)) Explain why hydrogen cyanide alone reacts extremely slowly with carbonyl compounds, and explain the role of adding sodium cyanide or sodium hydroxide as a catalyst. [3]

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- ((c)) State the geometry and bond angle around the carbonyl carbon in propanal, and explain why the addition product exists as an optically inactive racemic mixture. [3]

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- ((d)) Give the systematic IUPAC name of the 2-hydroxynitrile formed from propanal. [1]

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2 Addition of HCN to Symmetrical vs Unsymmetrical Carbonyls — 9701/21/O/N/23/Q3(a)–(c)

HARD · Syllabus 17.1

The addition of HCN to propanone and propanal produces different stereochemical results.

- ((a)) Write the balanced chemical equation for the reaction of propanone with HCN. [1]

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- ((b))** Draw the displayed formula of the organic product formed from propanone and state whether this product exhibits optical activity. Justify your answer. **[2]**

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- ((c))** State how the stereochemical outcome of adding HCN to propanal differs fundamentally from that of propanone. **[2]**

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3 Acid Hydrolysis of 2-Hydroxynitriles — 9701/22/F/M/22/Q3(a)–(c)

EASY · Syllabus 17.1

2-hydroxynitriles can be converted into 2-hydroxycarboxylic acids by heating under reflux with dilute acid.

- ((a))** 2-hydroxypropanenitrile, $\text{CH}_3\text{CH}(\text{OH})\text{CN}$, is heated under reflux with dilute hydrochloric acid. Write a balanced chemical equation for this reaction. **[2]**

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- ((b))** Give the systematic name and common name of the organic product formed. **[2]**

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- ((c))** Explain why the nitrile carbon ($-\text{CN}$) is converted to a carboxylic acid group while the alcohol group ($-\text{OH}$) remains unaffected during this reaction. **[2]**

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4 Reduction of Hydroxynitriles to Amino Alcohols — 9701/23/M/J/22/Q4(a)–(c)

HARD · Syllabus 17.1

2-hydroxynitriles can also be reduced to form 2-amino alcohols.

- ((a))** State the reagent and solvent required to reduce 2-hydroxypropanenitrile to 1-aminopropan-2-ol. **[2]**

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- ((b))** Write an equation for this reduction using $[\text{H}]$ to represent the reducing agent. **[1]**

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- ((c))** Explain why sodium borohydride, NaBH_4 , is NOT suitable for reducing a nitrile group, whereas lithium aluminium hydride, LiAlH_4 , is effective. **[2]**

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5 Comparative Reactivity of Aldehydes vs Ketones — 9701/21/O/N/22/Q3(a)–(c)

HARD · Syllabus 17.1

Aldehydes are generally significantly more reactive towards nucleophiles than ketones.

((a)) Explain this difference in reactivity in terms of the electronic (inductive) effects of alkyl groups. **[2]**

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((b)) Explain this difference in reactivity in terms of steric hindrance around the carbonyl carbon. **[2]**

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((c)) Rank the following three compounds in order of decreasing rate of nucleophilic addition of HCN: propanone, ethanal, methanal. Give a brief justification. **[2]**

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6 Carbon Chain Extension Strategy — 9701/22/M/J/21/Q4(a)–(c)

HARD · Syllabus 17.1

A synthetic chemist needs to synthesise 2-hydroxy-2-methylbutanoic acid starting from butanone.

((a)) State the reagent and catalyst needed for Step 1 to convert butanone into a 2-hydroxynitrile. **[2]**

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((b)) Draw the structural formula of the 2-hydroxynitrile intermediate formed in Step 1. **[1]**

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((c)) State the reagents, conditions, and write the chemical equation for Step 2 to convert this intermediate into 2-hydroxy-2-methylbutanoic acid. **[3]**

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7 Safety and Hazards of Cyanide in the Laboratory — 9701/11/M/J/23/Q20

EASY · Syllabus 17.1

Hydrogen cyanide, HCN, is an extremely toxic volatile liquid (b.p. 26 °C).

((a)) Explain why potassium cyanide, KCN, mixed with dilute acid is typically used in school laboratories rather than pure liquid HCN. **[2]**

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- ((b))** State two crucial laboratory safety precautions when preparing and using solutions containing cyanide ions. **[2]**
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8 Deducing Carbonyl Precursor from Hydrolysis Product — 9701/21/M/J/23/Q6

HARD · Syllabus 17.1

Compound P has molecular formula $C_5H_{10}O$. When reacted with $NaCN$ and dilute acid, it forms compound Q ($C_6H_{11}NO$). When Q is heated under reflux with dilute acid, compound R is obtained. Compound R is 2-hydroxy-3-methylbutanoic acid.

- ((a))** Draw the structural formula of compound R. **[1]**
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- ((b))** Deduce the structural formula and systematic IUPAC name of compound P. **[2]**
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- ((c))** State whether compound P is an aldehyde or a ketone. Justify your conclusion. **[2]**
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9 Optical Activity in Carbonyl Addition Products — 9701/22/O/N/23/Q6(a)-(c)

HARD · Syllabus 17.1

Consider the addition of HCN to ethanal.

- ((a))** Identify the chiral carbon atom in the product, 2-hydroxypropanenitrile, and state the four different groups attached to it. **[2]**
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- ((b))** Draw 3D wedge-and-dash diagrams showing the pair of optical isomers (enantiomers) of 2-hydroxypropanenitrile. **[2]**
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- ((c))** A polarimeter is used to measure the optical rotation of the reaction mixture after the addition is complete. Predict the angle of rotation and explain your answer. **[2]**
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10 Multiple Choice: Rate of Nucleophilic Addition — 9701/12/M/J/23/Q21

EASY · Syllabus 17.1

Which carbonyl compound reacts at the fastest rate with HCN in the presence of $NaCN$?

((a)) Choose the correct compound from the options below: [1]

- A Propan-2-one
B Butan-2-one
C Ethanal
D Pentan-3-one
- A
B
C
D

((b)) Explain why the compound you chose reacts faster than the other three. [2]

11 Addition of HCN to Aromatic Carbonyls — 9701/23/O/N/21/Q4(a)-(c)

HARD · Syllabus 17.1

Benzaldehyde, C_6H_5CHO , reacts with HCN in the presence of NaCN to form mandelonitrile.

((a)) Draw the structural formula of mandelonitrile. [1]

((b)) Explain why benzaldehyde reacts more slowly with HCN than ethanal, considering the delocalisation of electrons into the carbonyl group. [2]

((c)) Write the equation for the complete acid hydrolysis of mandelonitrile to mandelic acid (2-hydroxy-2-phenylethanoic acid). [1]

12 Role of pH in Cyanide Addition Kinetics — 9701/21/O/N/22/Q5(a)-(c)

HARD · Syllabus 17.1

The rate of addition of HCN to propanone varies dramatically with the pH of the reaction mixture.

((a)) Explain why the reaction is extremely slow at very low pH ($pH < 4$). [2]

((b)) Explain why the reaction is also slow at very high pH ($pH > 11$). [2]

- ((c)) Deduce the optimal pH range for this reaction and explain why this intermediate pH provides the maximum rate. [2]

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13 Addition of Bisulfite Ion (Extension Nucleophilic Addition) — 9701/22/M/J/22/Q5

HARD · Syllabus 17.1

Aldehydes and methyl ketones react with saturated aqueous sodium hydrogensulfite (sodium bisulfite, NaHSO₃) to form crystalline bisulfite addition adducts.

- ((a)) State the role of the hydrogensulfite ion (HSO₃⁻) in this reaction (electrophile or nucleophile). [1]

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- ((b)) Explain why bulky ketones such as pentan-3-one do not form bisulfite adducts, whereas ethanal and propanone react readily. [2]

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- ((c)) Describe how this reaction can be used practically to purify an aldehyde from a mixture of organic impurities. [2]

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14 Summary of Diagnostic Tests for Carbonyls — 9701/21/M/J/23/Q7(a)–(e)

HARD · Syllabus 17.1

Diagnostic reagents allow organic chemists to identify the presence of a carbonyl group and distinguish between aldehydes and ketones. Fig. 14.1 compares the four principal qualitative tests.

Diagnostic Chemical Tests for Carbonyl Compounds			
2,4-DNPH (Brady's Reagent)	Tollens' Reagent [Ag(NH ₃) ₂ ⁺]	Fehling's Solution (Cu ²⁺ in alkali)	Tri-iodomethane (I ₂ in NaOH)
- Tests for: Carbonyl (C=O) in both aldehydes & ketones - Reaction: Condensation (Addition-elimination) - Observation: Bright orange / yellow crystalline precipitate - Value: Filter, purify, measure m.p. of hydrazone to identify specific compound	- Tests for: Aldehydes (Ketones = negative) - Reaction: Mild oxidation $R-CHO + 2Ag^+ + 3OH^- \rightarrow R-COO^- + 2Ag(s) + 2H_2O$ - Observation: Silver mirror formed on clean test-tube walls (or grey precipitate)	- Tests for: Aldehydes (Ketones = negative) - Reaction: Mild oxidation $R-CHO + 2Cu^{2+} + 5OH^- \rightarrow R-COO^- + Cu_2O(s) + 3H_2O$ - Observation: Deep blue solution turns to brick-red precipitate of copper(I) oxide (Cu₂O)	- Tests for: Methyl carbonyl (CH₃-C(=O)- group) - Substrates: Ethanal and all methyl ketones (CH₃-CO-R) - Observation: Pale yellow crystalline precipitate of CHI₃ with antiseptic smell

Fig. 14.1: Summary of qualitative diagnostic tests: 2,4-DNPH, Tollens' reagent, Fehling's solution, and alkaline iodine.

- ((a)) State the observation when Brady's reagent (2,4-DNPH) is added to butanal, and describe how this test can be used to positively identify the specific aldehyde beyond just confirming a carbonyl group. [3]

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- ((b))** Describe how Tollens' reagent is prepared in the laboratory, state the visible observation with butanal, and write the ionic half-equation for the reduction of silver. **[3]**
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- ((c))** State the observation when Fehling's solution is heated with butanone compared to butanal, and identify the chemical formula of the precipitate formed with butanal. **[3]**
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- ((d))** Explain why ketones do not react with Tollens' reagent or Fehling's solution under ordinary laboratory conditions. **[2]**
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15 Condensation Mechanism with 2,4-DNPH — 9701/22/O/N/23/Q4(a)-(c)**HARD · Syllabus 17.1**

The reaction between a carbonyl compound and 2,4-dinitrophenylhydrazine is an addition-elimination (condensation) reaction.

- ((a))** Draw the structural formula of 2,4-dinitrophenylhydrazine. **[2]**
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- ((b))** Write the general word or chemical equation for the condensation of propanone with 2,4-DNPH, identifying the small molecule eliminated. **[2]**
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- ((c))** Explain why 2,4-DNPH reactions are typically carried out in the presence of an acid catalyst, such as dilute sulfuric acid. **[2]**
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16 Distinguishing Aldehydes and Ketones by Oxidation — 9701/21/O/N/21/Q3(a)-(c)**EASY · Syllabus 17.1**

Consider two unlabelled liquid samples: pentanal and pentan-3-one.

- ((a))** State the observation when each liquid is heated with acidified potassium dichromate(VI). **[2]**
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- ((b))** Write a balanced chemical equation for the oxidation of pentanal, using [O] to represent the oxidising agent. **[1]**
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- ((c)) Suggest why Tollens' reagent is preferable to acidified potassium dichromate(VI) when testing for the presence of an aldehyde in an unknown mixture that might also contain alcohols. [2]

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17 Redox Chemistry of Tollens' and Fehling's Tests — 9701/22/M/J/22/Q6(a)-(c)

HARD · Syllabus 17.1

In the Tollens' test, silver(I) complex ions are reduced, while in the Fehling's test, copper(II) tartrate complex ions are reduced.

- ((a)) Write the ionic equation for the oxidation of ethanal to ethanoate ions by Tollens' reagent in alkaline conditions. [2]

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- ((b)) State the oxidation number of copper before and after the reaction in Fehling's test, and write the formula of the precipitate. [2]

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- ((c)) Explain why Fehling's solution contains tartrate ions in addition to copper(II) sulfate and sodium hydroxide. [2]

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18 Aromatic Aldehydes and Fehling's Solution — 9701/23/M/J/23/Q4(a)-(b)

HARD · Syllabus 17.1

Benzaldehyde, C₆H₅CHO, gives a positive result with Tollens' reagent but does not react with Fehling's solution.

- ((a)) Describe the observation when benzaldehyde is warmed with Tollens' reagent. [1]

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- ((b)) Explain why aliphatic aldehydes (e.g. ethanal) react readily with Fehling's solution, whereas aromatic aldehydes (e.g. benzaldehyde) fail to react, referring to oxidising strength and conjugation. [2]

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19 Identifying Unknown Carbonyl Compound W — 9701/21/O/N/23/Q5

EASY · Syllabus 17.1

An organic liquid W has molecular formula C₄H₈O. It reacts with 2,4-DNPH to form a yellow-orange precipitate, but gives no reaction when warmed with Tollens' reagent.

- ((a)) State the conclusions that can be drawn from the positive 2,4-DNPH test and the negative Tollens' test. [2]

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((b)) Deduce the structural formula and systematic IUPAC name of compound W. **[2]**

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((c)) Predict the visible observation when compound W is tested with alkaline aqueous iodine. Justify your answer. **[2]**

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20**Purification and Characterisation by Melting Point — 9701/22/F/M/23/Q4(a)-(c)****HARD · Syllabus 17.1**

Table 20.1 lists the melting points of the 2,4-dinitrophenylhydrazone derivatives of several carbonyl compounds.

((a)) An unknown liquid has boiling point 56 °C. Suggest two carbonyl compounds that have similar boiling points. **[2]**

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((b)) The 2,4-DNPH derivative of this unknown liquid was prepared, recrystallised from ethanol, and melted sharply at 126 °C. **[3]**

Reference data states:

- Propanone 2,4-DNPH m.p. = 126 °C
- Propanal 2,4-DNPH m.p. = 154 °C

Identify the unknown liquid and explain why measuring the derivative's melting point is more reliable than measuring the liquid's boiling point.

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((c)) Explain why impurities in a recrystallised sample cause its melting point to be lower and melt over a wider temperature range. **[2]**

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21**Multiple Choice: Qualitative Carbonyl Tests — 9701/11/O/N/23/Q21****EASY · Syllabus 17.1**

Which reagent produces a precipitate with ethanal but NO precipitate with propanal?

((a)) Select the correct reagent: **[1]**

- A 2,4-dinitrophenylhydrazine
- B Tollens' reagent
- C Fehling's solution
- D Alkaline aqueous iodine
- A
- B
- C
- D

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((b)) Explain your choice. **[2]**

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22 Tollens' Test Practical Chemistry and Precautions — 9701/22/M/J/21/Q5

EASY · Syllabus 17.1

Tollens' reagent must always be prepared fresh and disposed of immediately after use.

((a)) Explain why Tollens' reagent must never be stored or allowed to dry out after an experiment. **[2]**

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((b)) State how used Tollens' reagent should be safely destroyed before disposal. **[1]**

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((c)) Why should a hot water bath be used to warm the mixture during the Tollens' test rather than direct heating with a Bunsen burner flame? **[1]**

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23 Oxidation Resistance of Cyclic Ketones — 9701/21/M/J/22/Q5(a)-(c)

HARD · Syllabus 17.1

Cyclohexanone is a cyclic ketone.

((a)) Draw the skeletal formula of cyclohexanone. **[1]**

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((b)) Predict the results (positive/negative and visible observation) when cyclohexanone is tested with: **[3]**

(i) 2,4-DNPH

(ii) Tollens' reagent

(iii) Alkaline aqueous iodine

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((c)) Suggest the product formed when cyclohexanone is reduced with NaBH₄. **[1]**

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24 Reaction of Carbonyls with Hydroxylamine — 9701/23/O/N/22/Q3(a)-(b)

HARD · Syllabus 17.1

Like 2,4-DNPH, hydroxylamine (H₂N-OH) reacts with carbonyl compounds by nucleophilic addition-elimination to form oximes.

((a)) Write the balanced chemical equation for the reaction of propanal with hydroxylamine. **[1]**

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((b)) Explain why the product oxime, CH₃CH₂CH=N-OH, exhibits geometric (cis-trans / E-Z) isomerism. **[2]**

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25 Distinguishing Three Isomeric Carbonyls — 9701/22/O/N/21/Q4

EASY · Syllabus 17.1

Three isomeric compounds X, Y, and Z have molecular formula C₅H₁₀O.

- X gives a silver mirror with Tollens' reagent and a yellow precipitate with alkaline aqueous iodine.
- Y gives a silver mirror with Tollens' reagent but no precipitate with alkaline aqueous iodine.
- Z gives no reaction with Tollens' reagent but gives a yellow precipitate with alkaline aqueous iodine.

((a))

Identify whether X, Y, and Z are aldehydes or ketones.

[2]

((b))

Deduce the structural formula and systematic name of compound Z.

[2]

((c))

Explain why compound X cannot exist as an aldehyde containing a methyl carbonyl group, or deduce its true identity.

[2]

26 Multiple Choice: Benedict's and Fehling's Tests — 9701/12/F/M/22/Q21

EASY · Syllabus 17.1

Benedict's solution and Fehling's solution both contain copper(II) ions. Which species is responsible for the red precipitate in a positive test?

((a))

Select the correct species:

- A CuO
- B Cu₂O
- C Cu(OH)₂
- D Cu

[1]

((b))

State the oxidation number of copper in this precipitate.

[1]

27 Reduction of Carbonyls with Sodium Borohydride — 9701/22/M/J/23/Q4(a)-(d)

HARD · Syllabus 17.1

Sodium borohydride, NaBH₄, is the standard laboratory reducing agent for aldehydes and ketones. Fig. 27.1 illustrates the reduction pathways to primary and secondary alcohols.

Reduction of Aldehydes & Ketones using Sodium Borohydride (NaBH₄)

Aldehyde Reduction -> Primary (1°) Alcohol	Ketone Reduction -> Secondary (2°) Alcohol
Reaction: $R-CHO + 2[H] \rightarrow R-CH_2OH$ Reagent & Conditions: • Sodium borohydride (NaBH₄) in aqueous ethanol at room temperature Role of Reagent: • Source of nucleophilic hydride ion (:H⁻) • Attacks electron-deficient carbonyl carbon • Followed by protonation from solvent (H₂O)	Reaction: $R_1-CO-R_2 + 2[H] \rightarrow R_1-CH(OH)-R_2$ Reagent & Conditions: • Sodium borohydride (NaBH₄) (safer, water-tolerant than LiAlH₄) Stereochemical Note: • Unsymmetrical ketones (R₁ ≠ R₂) form a new chiral carbon centre • Equal attack on either face yields a racemic mixture of enantiomers

Contrast: LiAlH₄ is a much stronger reducing agent used in dry ether (reacts violently with water).

Fig. 27.1: Reduction pathways of aldehydes and ketones using aqueous/ethanolic NaBH₄.

- ((a))** State the role of the BH_4 ion during the reduction and identify the nucleophilic species that attacks the carbonyl carbon. **[2]**
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- ((b))** Write a balanced chemical equation for the reduction of butanal to butan-1-ol using $[\text{H}]$ for the reducing agent. **[1]**
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- ((c))** Write a balanced chemical equation for the reduction of butanone to butan-2-ol using $[\text{H}]$. **[1]**
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- ((d))** Explain why reduction of butanone produces an optically inactive racemic mixture of butan-2-ol. **[3]**
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28 Comparison of NaBH_4 and LiAlH_4 — 9701/21/O/N/23/Q6(a)–(c)**HARD · Syllabus 17.1**

Both NaBH_4 and LiAlH_4 act as sources of hydride ions, but their reactivity differs significantly.

- ((a))** Explain why NaBH_4 can be safely dissolved in water or ethanol, whereas LiAlH_4 must be used in strictly anhydrous dry ether. **[2]**
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- ((b))** Write the equation for the violent reaction that occurs when LiAlH_4 comes into contact with water. **[1]**
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- ((c))** State which functional groups among aldehydes, ketones, and carboxylic acids can be reduced by: **[2]**
- (i) NaBH_4
- (ii) LiAlH_4
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29 Selective Reduction of Polyfunctional Compounds — 9701/22/O/N/22/Q5**HARD · Syllabus 17.1**

Compound G contains both an aldehyde group and a carbon-carbon double bond: $\text{CH}_3\text{-CH=CH-CHO}$ (but-2-enal).

- ((a))** Deduce the structure of the organic product formed when compound G reacts with NaBH_4 in aqueous ethanol. **[1]**
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((b)) Explain why NaBH_4 reduces the carbonyl group in compound G but leaves the $\text{C}=\text{C}$ double bond completely unaffected. **[2]**

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((c)) Suggest a reagent and catalyst that would reduce BOTH the $\text{C}=\text{C}$ double bond and the carbonyl group in compound G simultaneously. **[2]**

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30 Multi-Step Synthesis: Alcohol to Carboxylic Acid via Carbonyl — 9701/21/M/J/22/Q3

EASY · Syllabus 17.1

Outline a laboratory procedure for converting propan-1-ol into propanoic acid in high yield.

((a)) State the oxidising agent and conditions required for this transformation. **[2]**

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((b)) Write the balanced equation for the overall oxidation using $[\text{O}]$. **[1]**

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((c)) Explain why a vertical Liebig condenser (heating under reflux) is essential for this preparation. **[2]**

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31 Intermolecular Forces: Aldehydes vs Alcohols & Alkanes — 9701/23/M/J/22/Q2

EASY · Syllabus 17.1

Table 31.1 compares three compounds with relative molecular mass $M_r \approx 58-60$:

- Butane ($M_r = 58.1$, b.p. $-0.5\text{ }^\circ\text{C}$)
- Propanal ($M_r = 58.1$, b.p. $49\text{ }^\circ\text{C}$)
- Propan-1-ol ($M_r = 60.1$, b.p. $97\text{ }^\circ\text{C}$)

((a)) Identify the principal intermolecular forces present in pure liquid samples of each of the three compounds. **[3]**

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((b)) Explain why propanal has a substantially higher boiling point than butane. **[2]**

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((c)) Explain why propan-1-ol has a substantially higher boiling point than propanal. **[2]**

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32 Solubility of Carbonyls in Water — 9701/12/M/J/21/Q20

EASY · Syllabus 17.1

Short-chain aldehydes and ketones (e.g. methanal, ethanal, propanone) are completely miscible with water, but solubility decreases sharply as alkyl chain length increases.

- ((a))** Explain how propanone molecules are able to dissolve in water, despite being unable to form hydrogen bonds with one another in the pure liquid. [2]

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- ((b))** Draw a labelled diagram showing a hydrogen bond between a molecule of propanone and a molecule of water. [2]

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- ((c))** Explain why octan-2-one is completely insoluble in water. [1]

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33 Hydride Addition Mechanism for NaBH₄ — 9701/22/F/M/22/Q5(a)–(c)

HARD · Syllabus 17.1

The mechanism for the reduction of ethanal by NaBH₄ involves two steps.

- ((a))** Step 1: Draw curly arrows showing nucleophilic attack of a hydride ion (:H⁻) on ethanal to form an ethoxide intermediate. [2]

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- ((b))** Step 2: Show how the ethoxide intermediate is protonated by a water molecule from the solvent to yield ethanol and an OH⁻ ion. [2]

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- ((c))** State the change in hybridization of the carbonyl carbon atom as it transforms from ethanal to ethanol. [1]

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34 Multiple Choice: Reduction Products — 9701/11/M/J/22/Q23

EASY · Syllabus 17.1

Which compound is produced when 3-methylbutan-2-one is treated with NaBH₄ in aqueous ethanol?

- ((a))** Select the correct IUPAC name: [1]
- A 3-methylbutan-1-ol
 - B 3-methylbutan-2-ol
 - C 2-methylbutan-2-ol
 - D 2-methylbutane

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((b)) Write the structural formula of the product and identify any chiral centres. [2]

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35 Reversible Hydration of Carbonyls (Diols) — 9701/21/O/N/22/Q6

HARD · Syllabus 17.1

In aqueous solution, carbonyl compounds exist in dynamic equilibrium with their geminal diols (hydrates): $R_2C=O + H_2O \rightleftharpoons R_2C(OH)_2$.

((a)) In aqueous solution, methanal (HCHO) is over 99% converted into methanediol, whereas propanone is less than 0.1% converted into propane-2,2-diol. Explain this striking difference in equilibrium position. [3]

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((b)) Trichloroethanal (chloral, CCl_3CHO) forms a very stable crystalline solid hydrate, $CCl_3CH(OH)_2$ (chloral hydrate). Explain why the three chlorine atoms stabilise the hydrate form so dramatically. [2]

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36 Synthesis of Branched Alcohols via Carbonyl Addition — 9701/22/O/N/21/Q5

HARD · Syllabus 17.1

Aldehydes and ketones can be converted into carboxylic acids containing one additional carbon atom via hydroxynitrile intermediates.

((a)) Give the reaction sequence (reagents and conditions for each step) to convert ethanal into 2-hydroxypropanoic acid. [3]

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((b)) State how 2-hydroxypropanoic acid can be dehydrated to form an unsaturated carboxylic acid, and name the unsaturated acid. [2]

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37 Acidity of Alpha-Hydrogens in Carbonyls — 9701/23/M/J/21/Q4(a)-(c)

HARD · Syllabus 17.1

The hydrogen atoms attached to a carbon atom adjacent to a carbonyl group (alpha-hydrogens) are significantly more acidic than hydrogens in alkanes.

((a)) Explain why the alpha-protons of propanone can be removed by a strong base such as hydroxide or ethoxide ion. [2]

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((b)) Draw the resonance structures of the enolate anion formed when propanone loses an alpha-proton. **[2]**

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((c)) State the significance of enolate formation in the tri-iodomethane reaction. **[1]**

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38 Deducing Symmetrical Ketone from Mr and Reactions — 9701/21/M/J/23/Q4

EASY · Syllabus 17.1

A symmetrical ketone T has relative molecular mass $M_r = 86.1$. It gives an orange precipitate with 2,4-DNPH, but does NOT react with Tollens' reagent or alkaline aqueous iodine.

((a)) Deduce the molecular formula of ketone T. **[1]**

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((b)) Deduce the structural formula and systematic name of ketone T. **[2]**

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((c)) Draw the structure of the alcohol produced when ketone T is reduced with NaBH_4 . **[1]**

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39 Aldol Dimerisation (Extension of Carbonyl Addition) — 9701/22/M/J/23/Q8

HARD · Syllabus 17.1

When ethanal is treated with dilute aqueous sodium hydroxide at low temperatures, it undergoes a dimerisation reaction called the aldol addition.

((a)) Write the structural formula of the product, 3-hydroxybutanal (aldol). **[1]**

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((b)) Explain why this compound is named 'aldol'. **[1]**

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((c)) When 3-hydroxybutanal is warmed with dilute acid, it readily eliminates water. Deduce the structure and IUPAC name of the conjugated unsaturated product formed. **[2]**

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40 Mechanism of the Tri-iodomethane Cleavage — 9701/21/M/J/23/Q9(a)-(d)

HARD · Syllabus 17.1

The tri-iodomethane reaction is a hallmark diagnostic test for compounds containing the methyl carbonyl ($\text{CH}_3\text{-CO-}$) group. Fig. 40.1 illustrates the structural requirement, sequential chemical stages, and overall cleavage outcome.

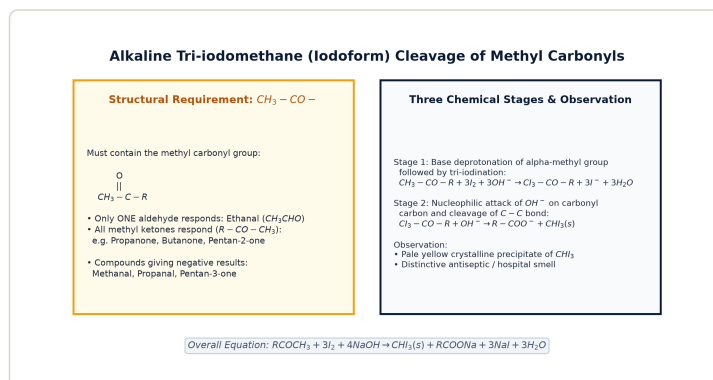


Fig. 40.1: Three chemical stages of the alkaline tri-iodomethane reaction on methyl ketones.

- ((a))** State the structural feature a carbonyl compound must possess to give a positive tri-iodomethane test, and name the ONLY aldehyde that gives this positive result. [2]

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- ((b))** Describe the reagents, conditions, and visible observation that confirm a positive test. [3]

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- ((c))** Write the balanced chemical equation for the reaction of propanone with iodine in aqueous sodium hydroxide. [2]

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- ((d))** Give the name and formula of the yellow precipitate and the name of the organic salt remaining in solution. [2]

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Distinguishing Isomeric Pentanones — 9701/22/O/N/23/Q3(a)-(c)

EASY · Syllabus 17.1

Pentan-2-one and pentan-3-one are positional isomers.

- ((a))** Draw the structural formulas of pentan-2-one and pentan-3-one. [2]

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- ((b))** State the reagent and condition that can distinguish between these two isomers. [2]

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- ((c))** State the observation for each isomer and identify the organic products formed with pentan-2-one. [3]

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42 Why Other Carbonyls Give Negative Iodoform Tests — 9701/21/O/N/22/Q7

HARD · Syllabus 17.1

Explain why the following compounds do NOT give a pale yellow precipitate with alkaline aqueous iodine.

((a)) Methanal, HCHO . [1]

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((b)) Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$. [2]

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((c)) Ethanoic acid, CH_3COOH (which contains a $\text{CH}_3\text{-C=O}$ group). [2]

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43 Infrared Spectroscopy of Carbonyl Compounds — 9701/22/M/J/22/Q7(a)–(c)

HARD · Syllabus 17.1

Infrared spectroscopy provides diagnostic evidence for carbonyl groups.

((a)) State the characteristic wave number range for the stretching vibration of the carbonyl (C=O) bond in aldehydes and ketones. [1]

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((b)) Explain why the C=O absorption peak in an infrared spectrum is extremely sharp and intense compared to C-C or C-H absorptions. [2]

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((c)) Aldehydes display a unique doublet of C-H stretching absorptions at $2700\text{--}2800\text{ cm}^{-1}$ (Fermi resonance). Explain how this feature allows an aldehyde to be distinguished from a ketone in an IR spectrum. [2]

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44 Mass Spectrometric Fragmentation of Carbonyls — 9701/23/M/J/23/Q6

HARD · Syllabus 17.1

The mass spectrum of butan-2-one ($M_r = 72$) exhibits major fragment peaks at $m/z = 57$, $m/z = 43$, and $m/z = 15$.

((a)) Identify the formula and structure of the molecular ion $[M]$. [1]

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((b)) Deduce the structure of the acylium fragment ion responsible for the base peak at $m/z = 43$. [1]

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((c)) Deduce the structure of the fragment ion responsible for the peak at $m/z = 57$. [1]

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((d)) Explain why cleavage occurs preferentially adjacent to the carbonyl group (alpha-cleavage). [2]

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45 Comprehensive Deduction of Unknown Carbonyl V — 9701/21/O/N/23/Q8

HARD · Syllabus 17.1

An organic compound V has composition 69.8% carbon, 11.6% hydrogen, and 18.6% oxygen by mass. Its mass spectrum shows a molecular ion peak at $m/z = 86$.

- Compound V produces an orange precipitate with 2,4-DNPH.
- Compound V produces a silver mirror when warmed with Tollens' reagent.
- Compound V has a branched carbon skeleton and displays four peaks in its carbon-13 NMR spectrum.

((a)) Determine the empirical formula and molecular formula of compound V. [2]

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((b)) State the functional group present in compound V and justify your conclusion from the chemical tests. [2]

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((c)) Deduce the structural formula and systematic name of compound V. [2]

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46 Multiple Choice: Chiral Centres in Carbonyl Addition — 9701/12/M/J/23/Q23

EASY · Syllabus 17.1

Which reaction produces an organic product that contains a chiral carbon atom?

((a)) Select the correct reaction: [1]

- Select the correct reaction:
- A Methanal + HCN / NaCN
- B Propanone + HCN / NaCN
- C Ethanal + HCN / NaCN
- D Pentan-3-one + NaBH₄
- A
- B
- C
- D

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((b)) Name the chiral product formed and identify its chiral carbon centre. [2]

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47 Multi-Step Synthetic Pathway: Alkene to Ketone — 9701/22/M/J/22/Q8

HARD · Syllabus 17.1

Devise a two-step synthetic pathway to convert propene into propanone.

- ((a))** Step 1: Hydrate propene to propan-2-ol. State the reagents and conditions, and explain why propan-2-ol is the major product. **[3]**

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- ((b))** Step 2: Oxidise propan-2-ol to propanone. State the reagents, conditions, and write the chemical equation using [O]. **[2]**

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- ((c))** Describe a test to confirm the presence of propanone in the final product mixture. **[2]**

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48 Multiple Choice: Fehling's vs Tollens' Reagents — 9701/11/O/N/22/Q22

EASY · Syllabus 17.1

When propanal is warmed with Fehling's solution, which species undergoes reduction?

- ((a))** Choose the correct species: **[1]**
- A Copper(II) ions
 - B Propanal
 - C Hydroxide ions
 - D Tartrate ions
- A
- B
- C
- D

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- ((b))** Write the ionic half-equation showing the reduction of this species. **[1]**

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49 Atmospheric Chemistry and Formaldehyde in Smog — 9701/21/M/J/21/Q6

HARD · Syllabus 17.1

Methanal (formaldehyde, HCHO) is a significant component of photochemical smog in urban atmospheres.

- ((a))** Methanal is formed by the incomplete combustion of hydrocarbons and the photo-oxidation of volatile organic compounds (VOCs). Write an equation for the complete combustion of methanal. **[1]**

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- ((b))** In the atmosphere, methanal absorbs UV sunlight ($\lambda < 330 \text{ nm}$) and undergoes photodissociation: $\text{HCHO} + h\nu \rightarrow \text{H}\cdot + \cdot\text{CHO}$. State the type of bond fission that occurs and define the term free radical. **[2]**

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- ((c)) Explain why methanal is a known health hazard and eye irritant.

[1]

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Comprehensive Chemical Roadmap of C3 Carbonyls — 9701/22/M/J/23/Q10

HARD · Syllabus 17.1

Propanal and propanone are structural isomers of formula C_3H_6O . Complete the comparison table below.

- ((a)) State the reagent, condition, and observation for a single test that gives a POSITIVE result with propanal but NO reaction with propanone.

[3]

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- ((b)) State the reagent, condition, and observation for a single test that gives a POSITIVE result with propanone but NO reaction with propanal.

[3]

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- ((c)) State the reagent and observation for a single test that gives a POSITIVE result with BOTH propanal and propanone.

[2]

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- ((d)) Draw the structural formula of the alcohol formed when propanal is reduced with $NaBH_4$ and when propanone is reduced with $NaBH_4$.

[2]

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Mark Scheme — Topic 17 — Carbonyl Compounds (Aldehydes & Ketones)

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Dipoles on carbonyl group: C(δ^+) and O(δ^-) in CH ₃ CH ₂ CHO [1] Curly arrow from lone pair on carbon of :CN to C(δ^+) of carbonyl [1] Curly arrow from C=O double bond onto oxygen atom forming tetrahedral alkoxide intermediate CH ₃ CH ₂ CH(CN)O [1] Curly arrow from lone pair on O to H of H-CN (or H ₂ O) regenerating :CN and forming 2-hydroxybutanenitrile [1]	4
1(b)	HCN is a very weak acid and only partially / weakly dissociates in aqueous solution, providing an extremely low concentration of nucleophilic :CN ions [1] Adding NaCN supplies a high concentration of free :CN ions [1] Adding NaOH deprotonates HCN (HCN + OH ⁻ → CN ⁻ + H ₂ O), generating the nucleophilic :CN required for the rate-determining step [1]	3
1(c)	Trigonal planar geometry with bond angle of 120° around the sp ² carbonyl carbon [1] The planar carbonyl centre can be attacked by the :CN nucleophile with equal probability from either the top face or the bottom face [1] This produces an equimolar (50:50) mixture of the two non-superimposable mirror-image enantiomers, cancelling optical rotation [1]	3
1(d)	2-hydroxybutanenitrile [1]	1
2(a)	CH ₃ COCH ₃ + HCN → (CH ₃) ₂ C(OH)CN [1]	1
2(b)	Displayed formula of 2-hydroxy-2-methylpropanenitrile: central carbon bonded to -OH, -CN, and two -CH ₃ groups [1] Optically inactive because the central carbon is achiral (it is bonded to two identical methyl groups) [1]	2
2(c)	In propanal, the product (2-hydroxybutanenitrile) possesses a chiral carbon bonded to four different groups (-H, -OH, -CN, -C ₂ H ₅) [1] It produces a pair of optical enantiomers (racemic mixture), whereas propanone produces an achiral product with no stereoisomers [1]	2
3(a)	CH ₃ CH(OH)CN + 2H ₂ O + HCl → CH ₃ CH(OH)COOH + NH ₄ Cl (or with H ⁺ : + 2H ₂ O + H ⁺ → CH ₃ CH(OH)COOH + NH ₄ ⁺) [2]	2
3(b)	Systematic name: 2-hydroxypropanoic acid [1] Common name: Lactic acid [1]	2
3(c)	The carbon in the nitrile group has a high partial positive charge due to the triple bond to electronegative nitrogen, making it susceptible to nucleophilic attack by water [1] The secondary alcohol -OH group is stable to dilute non-oxidising acid and does not undergo substitution or elimination under these mild reflux conditions [1]	2
4(a)	Reagent: Lithium aluminium hydride (LiAlH ₄) [1] Solvent: Dry ether (ethoxyethane) [1]	2
4(b)	CH ₃ CH(OH)CN + 4[H] → CH ₃ CH(OH)CH ₂ NH ₂ [1]	1
4(c)	NaBH ₄ is a milder reducing agent and lacks sufficient nucleophilic power / hydride transfer ability to attack the less polar CN triple bond [1] LiAlH ₄ contains a more polar, weaker Al-H bond that readily releases hydride ions (:H ⁻) with sufficient nucleophilicity to reduce nitriles [1]	2
5(a)	Alkyl groups exert an electron-releasing positive inductive (+I) effect [1] Ketones have two alkyl groups which reduce the partial positive charge on the carbonyl carbon (C(δ^+)) more than the single alkyl group in aldehydes, making the ketone carbon less electrophilic [1]	2
5(b)	Ketones possess two bulky alkyl groups attached to the carbonyl carbon, which crowd the reaction centre [1] This creates greater steric hindrance, obstructing the approach of the incoming nucleophile and destabilising the crowded tetrahedral intermediate [1]	2
5(c)	Methanal > ethanal > propanone [1] Methanal has zero alkyl groups (least steric crowding, most electrophilic carbon); propanone has two alkyl groups (most hindered, least electrophilic) [1]	2
6(a)	Reagents: HCN and NaCN (or KCN) catalyst (or NaCN + dilute H ₂ SO ₄) [1] Conditions: Room temperature / pH 8 [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
6(b)	$\text{CH}_3\text{-CH}_2\text{-C}(\text{CH}_3)(\text{OH})\text{-CN}$ [1]	1
6(c)	Reagents/conditions: Dilute hydrochloric acid (or dilute H_2SO_4), heat under reflux [1] Equation: $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)(\text{OH})\text{COOH} + \text{NH}_4^+$ [2]	3
7(a)	HCN is a volatile liquid that readily vaporises at room temperature to form deadly toxic gas [1] Solid KCN or NaCN can be weighed safely and dissolved in solution, generating HCN in situ under controlled, enclosed conditions [1]	2
7(b)	Perform all procedures strictly inside a well-ventilated fume cupboard / fume hood [1] Wear protective gloves and safety goggles, and keep an antidote / iron(II) sulfate deactivating solution readily available [1]	2
8(a)	$(\text{CH}_3)_2\text{CH-CH}(\text{OH})\text{-COOH}$ [1]	1
8(b)	Structural formula: $(\text{CH}_3)_2\text{CH-CHO}$ [1] Systematic name: 2-methylpropanal [1]	2
8(c)	Compound P is an aldehyde [1] In compound R, the carbon bearing the -OH group also has a hydrogen atom ($\text{CH}(\text{OH})$), showing that the carbonyl carbon had one H attached [1]	2
9(a)	Carbon-2 (C_2) is the chiral centre [1] Four groups: -H, - CH_3 , -OH, -CN [1]	2
9(b)	Correct 3D tetrahedral representations using wedges, dashes, and solid lines [1] Two non-superimposable mirror image structures drawn accurately [1]	2
9(c)	Angle of optical rotation = 0° / no optical rotation observed [1] A racemic mixture is formed containing exactly equal amounts of (+) and (-) enantiomers which rotate plane-polarised light by equal amounts in opposite directions, cancelling out [1]	2
10(a)	C (Ethanal) [1]	1
10(b)	Ethanal is the only aldehyde among the choices (the other three are ketones) [1] Aldehydes have only one electron-releasing alkyl group, giving the carbonyl carbon a higher positive charge (more electrophilic), and smaller steric hindrance to attack by CN [1]	2
11(a)	$\text{C}_6\text{H}_5\text{-CH}(\text{OH})\text{-CN}$ [1]	1
11(b)	The carbonyl group in benzaldehyde is conjugated with the pi electron cloud of the benzene ring [1] Delocalisation of pi electrons from the benzene ring into the carbonyl carbon reduces its partial positive charge (δ^+), making it less electrophilic than in aliphatic ethanal [1]	2
11(c)	$\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{C}_6\text{H}_5\text{CH}(\text{OH})\text{COOH} + \text{NH}_4^+$ [1]	1
12(a)	At very low pH (high $[\text{H}^+]$), the equilibrium $\text{HCN} \rightleftharpoons \text{H}^+ + \text{CN}^-$ is shifted far to the left [1] The concentration of nucleophilic $:\text{CN}^-$ ions is too low to attack the carbonyl carbon effectively in the rate-determining step [1]	2
12(b)	At very high pH (high $[\text{OH}^-]$), virtually all HCN is converted to CN^- , leaving negligible undissociated HCN molecules in solution [1] Undissociated HCN is needed in Step 2 to protonate the alkoxide intermediate and complete the reaction while regenerating the catalyst [1]	2
12(c)	Optimal pH is approximately 7.5 to 8.5 (weakly alkaline) [1] This provides a sufficient concentration of both the attacking nucleophile ($:\text{CN}^-$) and the proton-donating species (HCN) [1]	2
13(a)	Nucleophile (attacks via lone pair on sulfur) [1]	1
13(b)	The bisulfite ion is a very bulky nucleophile [1] Pentan-3-one has two large ethyl groups attached to the carbonyl carbon which create severe steric crowding, preventing the bisulfite adduct from forming [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
13(c)	Shake impure aldehyde with saturated NaHSO ₃ ; the solid crystalline adduct precipitates out and is filtered off, washing away impurities [1] Add dilute acid or dilute alkali to decompose the adduct, regenerating the pure aldehyde [1]	2
14(a)	Orange / yellow crystalline precipitate formed [1] Filter and recrystallise the 2,4-dinitrophenylhydrazone derivative [1] Dry and determine its sharp melting point, then compare this value with reference tables of known melting points [1]	3
14(b)	Add dilute aqueous ammonia dropwise to aqueous silver nitrate (AgNO ₃) until the initial brown precipitate of Ag ₂ O just dissolves to form colourless [Ag(NH ₃) ₂] [1] Silver mirror forms on test tube walls (or black precipitate) [1] [Ag(NH ₃) ₂] + e ⁻ → Ag(s) + 2NH ₃ (or Ag + e ⁻ → Ag) [1]	3
14(c)	With butanone: No reaction / remains clear deep blue [1] With butanal: Blue solution produces a brick-red / orange-red precipitate [1] Precipitate formula: Cu ₂ O / copper(I) oxide [1]	3
14(d)	Oxidation of a carbonyl carbon requires the breaking of a C-H bond attached directly to the carbonyl group [1] Ketones have only strong C-C bonds attached to the carbonyl carbon, which cannot be oxidised by mild oxidising agents like Tollens' or Fehling's [1]	2
15(a)	Benzene ring with -NH-NH ₂ group attached [1] Two nitro groups (-NO ₂) at positions 2 and 4 relative to the hydrazine group [1]	2
15(b)	(CH ₃) ₂ C=O + H ₂ N-NH-C ₆ H ₃ (NO ₂) ₂ → (CH ₃) ₂ C=N-NH-C ₆ H ₃ (NO ₂) ₂ + H ₂ O [1] Small molecule eliminated: Water (H ₂ O) [1]	2
15(c)	The acid protonates the carbonyl oxygen atom, creating a positive charge on oxygen [1] This significantly increases the partial positive charge on the carbonyl carbon (increases electrophilicity), making it more susceptible to nucleophilic attack by the weakly basic hydrazine nitrogen [1]	2
16(a)	Pentanal: Orange solution turns green [1] Pentan-3-one: Solution remains orange / no colour change [1]	2
16(b)	CH ₃ CH ₂ CH ₂ CH ₂ CHO + [O] → CH ₃ CH ₂ CH ₂ CH ₂ COOH [1]	1
16(c)	Acidified dichromate(VI) oxidises both primary and secondary alcohols (turning green), leading to a false positive for an aldehyde [1] Tollens' reagent is a very mild oxidising agent that oxidises aldehydes specifically without reacting with primary or secondary alcohols [1]	2
17(a)	CH ₃ CHO + 2[Ag(NH ₃) ₂] + 3OH ⁻ → CH ₃ COO ⁻ + 2Ag + 4NH ₃ + 2H ₂ O [2]	2
17(b)	Copper oxidation number changes from +2 to +1 [1] Precipitate formula: Cu ₂ O [1]	2
17(c)	Tartrate ions act as bidentate/polydentate chelating ligands that complex Cu ²⁺ ions [1] This prevents Cu ²⁺ from precipitating out as insoluble copper(II) hydroxide, Cu(OH) ₂ , in the strongly alkaline solution [1]	2
18(a)	Silver mirror forms on inside of the test tube / grey precipitate of silver [1]	1
18(b)	Fehling's solution is a weaker oxidising agent than Tollens' reagent [1] Conjugation of the carbonyl group with the benzene ring delocalises electron density and stabilises the aldehyde carbonyl towards oxidation, making benzaldehyde harder to oxidise [1]	2
19(a)	Positive 2,4-DNPH confirms presence of a carbonyl group (C=O) [1] Negative Tollens' test confirms W is a ketone (not an aldehyde) [1]	2
19(b)	Structural formula: CH ₃ -CO-CH ₂ -CH ₃ (or CH ₃ COCH ₂ CH ₃) [1] Systematic name: Butanone [1]	2
19(c)	Pale yellow precipitate forms [1] Butanone contains the methyl carbonyl group (CH ₃ -C=O) required for the positive tri-iodomethane reaction [1]	2
20(a)	Propanone (b.p. 56 °C) and propanal (b.p. 49 °C) [2]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
20(b)	Unknown is propanone [1] Solid derivatives can be purified to very high purity by recrystallisation [1] Melting points of solid derivatives are sharp and differ widely (126 °C vs 154 °C), whereas boiling points of liquids are sensitive to atmospheric pressure and close together [1]	3
20(c)	Impurities disrupt the regular crystal lattice structure, weakening intermolecular attractions [1] Less thermal energy is required to disrupt the lattice (lowering m.p.), and different regions of the non-uniform lattice melt at different temperatures (broadening range) [1]	2
21(a)	D (Alkaline aqueous iodine) [1]	1
21(b)	Ethanal is the only aldehyde with a CH ₃ -C=O group, so it gives a pale yellow precipitate of CHI ₃ [1] Propanal has a CH ₃ CH ₂ -C=O group (not a methyl carbonyl) so it gives a negative result with alkaline iodine, whereas 2,4-DNPH, Tollens', and Fehling's react with both [1]	2
22(a)	On standing or drying, it forms explosive silver fulminate / silver nitride (Ag ₃ N) [1] This precipitate is shock-sensitive and can detonate violently on friction or agitation [1]	2
22(b)	Acidify with dilute nitric acid (or dilute hydrochloric acid) to dissolve silver compounds before disposal [1]	1
22(c)	Aldehydes are highly volatile and flammable; gentle warming in a water bath prevents boiling over, charring, and igniting flammable vapours [1]	1
23(a)	Six-membered ring with a =O on one vertex [1]	1
23(b)	(i) Positive: Orange / yellow precipitate [1] (ii) Negative: No silver mirror / solution remains colourless [1] (iii) Negative: No pale yellow precipitate / no reaction (not a methyl ketone) [1]	3
23(c)	Cyclohexanol [1]	1
24(a)	CH ₃ CH ₂ CHO + H ₂ NOH → CH ₃ CH ₂ CH=N-OH + H ₂ O [1]	1
24(b)	Restricted rotation about the C=N double bond [1] Both the carbon and nitrogen have two different groups attached (carbon has -H and -C ₂ H ₅ ; nitrogen has -OH and a lone pair) [1]	2
25(a)	X and Y are aldehydes (positive Tollens'); Z is a ketone (negative Tollens') [2]	2
25(b)	Z is a methyl ketone with 5 carbons: pentan-2-one (CH ₃ COCH ₂ CH ₂ CH ₃) or 3-methylbutan-2-one (CH ₃ COCH(CH ₃) ₂) [2]	2
25(c)	Ethanal is the ONLY aldehyde that gives a positive iodoform test (having CH ₃ -C=O); no 5-carbon aldehyde can contain a methyl carbonyl group, so observation for X would indicate an impurity or invalid premise [2]	2
26(a)	B (Cu ₂ O) [1]	1
26(b)	+1 [1]	1
27(a)	BH ₄ acts as a source of hydride ions (:H) / reducing agent [1] The nucleophile is the hydride ion (:H) [1]	2
27(b)	CH ₃ CH ₂ CH ₂ CHO + 2[H] → CH ₃ CH ₂ CH ₂ CH ₂ OH [1]	1
27(c)	CH ₃ COCH ₂ CH ₃ + 2[H] → CH ₃ CH(OH)CH ₂ CH ₃ [1]	1
27(d)	The carbonyl group in butanone is planar around the sp ² carbon atom [1] The hydride ion (:H) has an equal probability of attacking from the top face or bottom face of the planar ketone [1] This produces an equimolar (50:50) mixture of the two optical enantiomers of butan-2-ol, resulting in zero net optical rotation (racemate) [1]	3
28(a)	The B-H bond in NaBH ₄ is relatively covalent and less polarised, reacting only very slowly with water at room temperature [1] The Al-H bond in LiAlH ₄ is much more ionic/polar and reacts violently/exothermically with water to release flammable hydrogen gas [1]	2
28(b)	LiAlH ₄ + 4H ₂ O → LiOH + Al(OH) ₃ + 4H ₂ [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
28(c)	(i) NaBH ₄ reduces aldehydes and ketones only (cannot reduce carboxylic acids) [1] (ii) LiAlH ₄ reduces aldehydes, ketones, AND carboxylic acids (to 1° alcohols) [1]	2
29(a)	CH ₃ -CH=CH-CH ₂ OH (but-2-en-1-ol) [1]	1
29(b)	NaBH ₄ reduces via nucleophilic attack of hydride ion (:H) on an electron-deficient carbon [1] The carbonyl carbon is electron-deficient (C(δ ⁺)), whereas the C=C double bond is electron-rich (pi bond cloud) and repels the negatively charged hydride nucleophile [1]	2
29(c)	Reagent: Hydrogen gas (H ₂) [1] Catalyst: Nickel (Ni) heated or Platinum (Pt) / Palladium (Pd) at room temperature [1]	2
30(a)	Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄) in excess [1] Heat under reflux [1]	2
30(b)	CH ₃ CH ₂ CH ₂ OH + 2[O] → CH ₃ CH ₂ COOH + H ₂ O [1]	1
30(c)	The intermediate propanal has a low boiling point (49 °C) and vaporises rapidly [1] Reflux condenses volatile propanal vapours and returns them to the hot flask, ensuring complete oxidation to propanoic acid rather than loss of intermediate [1]	2
31(a)	Butane: London dispersion (induced dipole-dipole) forces only [1] Propanal: Permanent dipole-dipole forces and London dispersion forces [1] Propan-1-ol: Intermolecular hydrogen bonding, permanent dipole-dipole, and London dispersion forces [1]	3
31(b)	Propanal contains a polar carbonyl group (C=O) giving permanent dipoles [1] Permanent dipole-dipole forces are stronger than the weak dispersion forces in non-polar butane, requiring more thermal energy to separate molecules [1]	2
31(c)	Propan-1-ol has an -OH group and forms intermolecular hydrogen bonds with neighbouring molecules [1] Hydrogen bonds are significantly stronger than permanent dipole-dipole forces in propanal, requiring much more thermal energy to vaporise [1]	2
32(a)	Propanone has lone pairs on its electronegative carbonyl oxygen atom [1] It can act as a hydrogen-bond acceptor, forming hydrogen bonds with the δ ⁺ hydrogen atoms of water molecules [1]	2
32(b)	Correct dipoles shown on water (H(δ ⁺), O(δ ⁻)) and propanone (O(δ ⁻)) [1] Dashed line from lone pair on propanone oxygen to H(δ ⁺) of water with ~180° bond angle [1]	2
32(c)	The large non-polar hydrophobic hydrocarbon chain disrupts water's extensive hydrogen bond network without generating sufficient solvation energy to overcome dispersion forces [1]	1
33(a)	Curly arrow from lone pair on :H to carbonyl carbon C(δ ⁺) of CH ₃ CHO [1] Curly arrow from C=O double bond onto carbonyl oxygen forming CH ₃ CH ₂ O [1]	2
33(b)	Curly arrow from lone pair on oxygen of CH ₃ CH ₂ O to H of H-OH [1] Curly arrow breaking H-OH bond onto oxygen to release OH [1]	2
33(c)	From sp ² (trigonal planar, 120°) to sp ³ (tetrahedral, 109.5°) [1]	1
34(a)	B (3-methylbutan-2-ol) [1]	1
34(b)	(CH ₃) ₂ CH-CH(OH)-CH ₃ [1] Carbon-2 is a chiral centre (bonded to -H, -OH, -CH ₃ , -CH(CH ₃) ₂) [1]	2
35(a)	Steric factor: Methanal has two tiny H atoms; converting from planar 120° to tetrahedral 109.5° causes no steric crowding, whereas propanone has two bulky methyl groups which crowd each other in the tetrahedral hydrate [1] Electronic factor: In propanone, two electron-releasing methyl groups stabilise the planar C=O bond, reducing the driving force for addition [1] Methanal has no alkyl groups, leaving the carbonyl carbon highly electron-deficient and reactive towards water addition [1]	3
35(b)	The three strongly electronegative chlorine atoms exert a powerful electron-withdrawing negative inductive (-I) effect [1] This makes the carbonyl carbon exceptionally electron-deficient in chloral, shifting the hydration equilibrium heavily towards the gem-diol [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
36(a)	Step 1: React ethanal with HCN + NaCN (or KCN) at room temp / pH 8 to form 2-hydroxypropanenitrile [2] Step 2: Heat under reflux with dilute hydrochloric acid (dilute HCl(aq)) [1]	3
36(b)	Heat with concentrated sulfuric acid (conc. H ₂ SO ₄) or concentrated phosphoric acid [1] Name: Prop-2-enoic acid (acrylic acid) [1]	2
37(a)	The strongly electronegative carbonyl oxygen pulls electron density towards itself (polar C=O), weakening the adjacent C-H bonds [1] The resulting carbanion conjugate base is stabilised by resonance delocalisation of negative charge onto the electronegative oxygen atom [1]	2
37(b)	CH ₂ -C(=O)-CH ₃ ↔ CH ₂ =C(O)-CH ₃ [2]	2
37(c)	Enolate formation allows the nucleophilic alpha-carbon to attack iodine molecules, enabling successive halogenation to Cl ₃ -CO-R [1]	1
38(a)	C ₅ H ₁₀ O (5 x 12 + 10 x 1 + 16 = 86) [1]	1
38(b)	Structural formula: CH ₃ -CH ₂ -CO-CH ₂ -CH ₃ [1] Systematic name: Pentan-3-one (symmetrical, negative iodoform confirms not pentan-2-one) [1]	2
38(c)	CH ₃ -CH ₂ -CH(OH)-CH ₂ -CH ₃ (pentan-3-ol) [1]	1
39(a)	CH ₃ -CH(OH)-CH ₂ -CHO [1]	1
39(b)	It contains both an aldehyde ('ald-') and an alcohol ('-ol') functional group in the same molecule [1]	1
39(c)	Structure: CH ₃ -CH=CH-CHO [1] Systematic name: But-2-enal [1]	2
40(a)	Must contain the methyl carbonyl group: CH ₃ -C(=O)- [1] Ethanal (acetaldehyde) [1]	2
40(b)	Reagents: Aqueous iodine (I ₂) and aqueous sodium hydroxide (NaOH) (or KI + NaOCl) [1] Condition: Warm gently / 50-60 °C [1] Observation: Pale yellow crystalline precipitate with a characteristic medicinal / antiseptic odour [1]	3
40(c)	CH ₃ COCH ₃ + 3I ₂ + 4NaOH → CHI ₃ (s) + CH ₃ COONa + 3NaI + 3H ₂ O [2]	2
40(d)	Precipitate: Tri-iodomethane, CHI ₃ [1] Salt: Sodium ethanoate (CH ₃ COONa) [1]	2
41(a)	Pentan-2-one: CH ₃ -CO-CH ₂ -CH ₂ -CH ₃ [1] Pentan-3-one: CH ₃ -CH ₂ -CO-CH ₂ -CH ₃ [1]	2
41(b)	Reagent: Aqueous iodine in aqueous sodium hydroxide (I ₂ / NaOH(aq)) [1] Condition: Warm gently [1]	2
41(c)	Pentan-2-one: Pale yellow crystalline precipitate formed [1] Pentan-3-one: No precipitate / solution remains brown or pale yellow [1] Organic products from pentan-2-one: Tri-iodomethane (CHI ₃) and sodium butanoate (CH ₃ CH ₂ CH ₂ COONa) [1]	3
42(a)	Methanal has no methyl group (no alpha-hydrogens at all) [1]	1
42(b)	Propanal has an ethyl group attached to the carbonyl (CH ₃ CH ₂ -CO-), with only two alpha-hydrogens (a -CH ₂ - group) [1] It cannot undergo tri-iodination to form the -Cl ₃ leaving group essential for cleavage to CHI ₃ [1]	2
42(c)	In alkaline solution, ethanoic acid is immediately deprotonated to the resonance-stabilised ethanoate anion, CH ₃ COO ⁻ [1] The negative charge on the carboxylate group strongly repels incoming hydroxide nucleophiles and prevents nucleophilic cleavage [1]	2
43(a)	1670 - 1740 cm ⁻¹ (or 1680 - 1750 cm ⁻¹) [1]	1
43(b)	The C=O bond has a very large permanent dipole moment due to the large electronegativity difference between carbon and oxygen [1] Vibration of this bond produces a large change in dipole moment, giving a very strong absorption band according to selection rules [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
43(c)	The aldehyde C-H bond attached directly to the carbonyl carbon produces characteristic twin bands at 2720 cm ⁻¹ and 2820 cm ⁻¹ [1] Ketones lack an aldehyde C-H bond and do not show these twin absorptions, showing only standard aliphatic C-H peaks above 2900 cm ⁻¹ [1]	2
44(a)	[CH ₃ COCH ₂ CH ₃] ⁺ at m/z = 72 [1]	1
44(b)	[CH ₃ -CO] (acetyl / ethanoyl cation, m/z = 15 + 28 = 43) [1]	1
44(c)	[CH ₃ CH ₂ -CO] (propanoyl cation, m/z = 29 + 28 = 57) OR [CH ₃ CH ₂ CH ₂ CH ₂] ⁺ [1]	1
44(d)	Alpha-cleavage generates resonance-stabilised acylium ions: R-C=O <-> R-CO ⁺ [1] The octet rule is satisfied for all atoms in the R-CO resonance contributor, making acylium ions unusually stable [1]	2
45(a)	Moles: C = 69.8/12 = 5.817; H = 11.6/1.0 = 11.6; O = 18.6/16 = 1.163 [1] Ratio C:H:O = 5:10:1 -> Empirical and Molecular formula = C ₅ H ₁₀ O (Mr = 86) [1]	2
45(b)	Aldehyde functional group (-CHO) [1] Positive 2,4-DNPH confirms carbonyl group; positive Tollens' test confirms aldehyde (not ketone) [1]	2
45(c)	Structural formula: (CH ₃) ₂ CH-CH ₂ -CHO (3-methylbutanal) OR (CH ₃) ₃ C-CHO (2,2-dimethylpropanal) [1] 4 carbon environments fits 3-methylbutanal (two equivalent methyls, CH, CH ₂ , CHO); systematic name: 3-methylbutanal [1]	2
46(a)	C (Ethanal + HCN / NaCN) [1]	1
46(b)	2-hydroxypropanenitrile [1] Carbon-2 is chiral, bonded to -H, -CH ₃ , -OH, and -CN [1]	2
47(a)	Reagents/conditions: Steam (H ₂ O(g)) and concentrated H ₃ PO ₄ at 300 °C and 60 atm (or conc. H ₂ SO ₄ then H ₂ O) [1] Electrophilic addition proceeds via the more stable secondary carbocation intermediate (CH ₃ -CH-CH ₃) rather than primary [1] Secondary carbocation is stabilised by the inductive electron-releasing effect of two methyl groups (Markovnikov's rule) [1]	3
47(b)	Acidified potassium dichromate(VI) (K ₂ Cr ₂ O ₇ + dilute H ₂ SO ₄), heat under reflux [1] CH ₃ CH(OH)CH ₃ + [O] -> CH ₃ COCH ₃ + H ₂ O [1]	2
47(c)	Add 2,4-DNPH: orange precipitate confirms carbonyl; negative Tollens' test confirms ketone [1] OR: warm with alkaline aqueous iodine to observe a pale yellow precipitate of CHI ₃ [1]	2
48(a)	A (Copper(II) ions) [1]	1
48(b)	2Cu ²⁺ + 2OH ⁻ + 2e ⁻ -> Cu ₂ O + H ₂ O (or Cu ²⁺ + e ⁻ -> Cu) [1]	1
49(a)	HCHO + O ₂ -> CO ₂ + H ₂ O [1]	1
49(b)	Homolytic fission [1] Free radical: A species containing an unpaired valence electron [1]	2
49(c)	Toxic, carcinogenic, cross-links biological proteins, and causes acute mucosal / respiratory irritation [1]	1
50(a)	Tollens' reagent (warm gently) [1] Propanal: Silver mirror formed on test tube wall [1] Propanone: No reaction / solution remains colourless (OR Fehling's: propanal turns brick-red, propanone stays blue) [1]	3
50(b)	Alkaline aqueous iodine (I ₂ + NaOH(aq), warm gently) [1] Propanone: Pale yellow crystalline precipitate of CHI ₃ [1] Propanal: No precipitate / negative result [1]	3
50(c)	2,4-dinitrophenylhydrazine (Brady's reagent / 2,4-DNPH) [1] Both form a bright orange / yellow crystalline precipitate [1]	2
50(d)	From propanal: CH ₃ CH ₂ CH ₂ OH (propan-1-ol, 1° alcohol) [1] From propanone: CH ₃ CH(OH)CH ₃ (propan-2-ol, 2° alcohol) [1]	2